IH Used Paper BTECH-28-10-20

Code No: 135CU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, October - 2020
NON-CONVENTIONAL POWER GENERATION
(Common to CE, ME, ECE, CSE, MCT)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the working of concentrated collector with the help of a neat sketch.
b) How solar radiation effect on tilted surface. [7+8]

2.a) Briefly explain the need for solar energy collection. Differentiate between flat plate collectors and concentrating collectors.
b) Explain any one solar radiation measuring instrument with a neat sketch. [7+8]

3.a) Explain the method of solar thermal energy storage using sensible heat.
b) Explain solar photovoltaic energy conversion with graph. [7+8]

4.a) What is MPPT? Describe the construction and working of MPPT system.
b) Draw the I-V characteristics of PV cell. [7+8]

5.a) Briefly explain the Indian wind energy potential.
b) Explain the working of horizontal axis wind mill with a neat sketch. [7+8]

6.a) Discuss about the classification and applications of wind energy conversion systems.
b) Discuss the factors influencing wind shear and turbulence. [7+8]

7.a) Enumerate the prime movers used in geothermal power stations.
b) Explain about the storage and handling of biomass. [7+8]

8.a) Define fuel cell. Explain in detail about the working of fuel cell.
b) List out the limitations of storage batteries. [7+8]

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